

Kundan Krishna

Research Scientist, Apple Inc.

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Research Interests

- **General:** Foundation models, Natural language processing, AI safety
- **Focus areas:** Automated redteaming of foundation models, Efficient guardrails and mitigations for AI safety issues, Synthetic data generation, Ensuring factual correctness of LLM-generated text

Education

PhD in Language and Information Technologies

August 2018 - May 2024

CARNEGIE MELLON UNIVERSITY

Pittsburgh, PA

- Thesis: Improving the reliability of language models for summarization
- Contribution: Advanced the state of abstractive summarization by designing noise-robust systems, synthetic pre-training methodologies, and automated fact-checking tools.
- Advisors: [Zachary Lipton](#), [Jeffrey Bigham](#)
- Committee members: [Sherry Wu](#), [Byron Wallace](#), [Alexander Rush](#)
- CGPA: 3.96/4.0

B.Tech. in Computer Science and Engineering

July 2012 - June 2016

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Kanpur, India

- CPI: 9.9/10

Professional Experience

Foundation Model/LLM Research Scientist

June 2024 - present

APPLE AIML

Pittsburgh, PA

- Working on research problems focused on tackling AI safety issues and alignment, specifically focusing on efficient intervention and automated vulnerability discovery.

Research Intern

May 2023 - Sep 2023

ABRIDGE AI

Pittsburgh, PA

- Designed an interactive system to help doctors to easily augment LLM-generated summaries of patient visits.

Research Intern

June 2022 - Aug 2022

GOOGLE BRAIN

Mountain View, CA

- Worked on improving robustness of summarization models to the presence of noise in the input.
- Designed an algorithm to detect and remove arbitrary noise types without prior knowledge of its characteristics.

Applied Scientist Intern

May 2020 - Aug 2020

AMAZON ALEXA AI

Pittsburgh, PA (remote)

- Worked on improving entity linking for video entities requested via Alexa.
- Implemented and improved a vector search based method delivering over 20% improvement in accuracy.

Research Engineer

June 2016 - July 2018

ADOBE RESEARCH, INDIA

Bangalore, India

- Identified potential applications of machine learning in Adobe's products.
- Designed models and algorithms and implemented research prototypes.

Selected Publications

- **Disentangled Safety Adapters Enable Efficient Guardrails and Flexible Inference-Time Alignment.**
Kundan Krishna, Joseph Y Cheng, Charles Maalouf, Leon A Gatys. [arXiv:2506.00166](https://arxiv.org/abs/2506.00166) (preprint)

- **GenAudit: Fixing Factual Errors in Language Model Outputs with Evidence.** Kundan Krishna, Sanjana Ramprasad, Prakhar Gupta, Byron Wallace, Zachary C Lipton, Jeffrey P Bigham. *Neurips Safe Generative AI Workshop 2024.*
- **USB: A Unified Summarization Benchmark Across Tasks and Domains.** Kundan Krishna, Prakhar Gupta, Sanjana Ramprasad, Byron C Wallace, Jeffrey P Bigham, Zachary C Lipton. *Conference on Empirical Methods in Natural Language Processing (EMNLP): Findings, 2023*
- **Downstream Datasets Make Surprisingly Good Pretraining Corpora.** Kundan Krishna, Saurabh Garg, Jeffrey Bigham, Zachary Lipton. *Annual Meeting of the Association for Computational Linguistics (ACL), 2023*
- **Improving the Robustness of Summarization Models by Detecting and Removing Input Noise.** Kundan Krishna, Yao Zhao, Jie Ren, Balaji Lakshminarayanan, Jiaming Luo, Mohammad Saleh, Peter J Liu. *Conference on Empirical Methods in Natural Language Processing (EMNLP): Findings, 2023*
- **Out-of-Distribution Detection and Selective Generation for Conditional Language Models.** Jie Ren, Jiaming Luo, Yao Zhao, Kundan Krishna, Mohammad Saleh, Balaji Lakshminarayanan, Peter J Liu. *International Conference on Learning Representations (ICLR), 2023*
- **Does Pretraining for Summarization Require Knowledge Transfer?.** Kundan Krishna, Jeffrey Bigham, Zachary Lipton. *Conference on Empirical Methods in Natural Language Processing (EMNLP): Findings, 2021*
- **Generating SOAP Notes from Doctor-Patient Conversations Using Modular Summarization Techniques.** Kundan Krishna, Sopan Khosla, Jeffrey Bigham, Zachary Lipton. *Annual Meeting of the Association for Computational Linguistics (ACL), 2021*
- **Extracting Structured Data from Physician-Patient Conversations By Predicting Noteworthy Utterances.** Kundan Krishna, Amy Pavel, Benjamin Schloss, Jeffrey Bigham, Zachary Lipton. *International Workshop on Health Intelligence at AAAI 2020*
- **Generating topic-oriented summaries using neural attention.** Kundan Krishna, Balaji V. Srinivasan. *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2018*
- **Vocabulary Tailored Summary Generation.** Kundan Krishna, Aniket Murhekar, Saumitra Sharma, Balaji V. Srinivasan. *International Conference on Computational Linguistics (COLING), 2018*
- **An LSTM based method for prediction of human activities with durations.** Kundan Krishna*, Deepali Jain*, Sanket Mehta, Sunav Choudhary. *UbiComp 2018*

Patents

- **Generating a topic-based summary of textual content.** *US Patent 10685050*
- **Bundling online content fragments for presentation based on content-specific metrics and inter-content constraints.** *US Patent 10891667*
- **Constructing content based on multi-sentence compression of source content.** *US Patent 10949452*
- **Visualizing natural language through 3D scenes in augmented reality.** *US Patent 10665030*
- **Generating a Targeted Summary of Textual Content Tuned to a Target Audience Vocabulary.** *US Patent 10534854*
- **Methods of automatically generating formatted annotations of doctor-patient conversations.** *US Patent App. 17736624*

Honors & Awards

2014-2016	Academic Excellence Award for 3 consecutive years for outstanding academic performance	IIT Kanpur
2014	Dr. Elizabeth and Varkey Cherian Award for best research project in the Summer Undergraduate Research Grant for Excellence(SURGE) program (jointly awarded to 2 out of 43 total research projects)	IIT Kanpur